

Deterministic Policies for Constrained Reinforcement Learning in Polynomial Time

Jeremy McMahan



Problem

For any cost criteria C , we wish to solve:

$$\max_{\pi \in \Pi} \mathbb{E}_M^\pi \left[\sum_{h=1}^H r_h(s_h, a_h) \right] \quad \text{s.t.} \quad \begin{cases} C_M^\pi \leq B \\ \pi \text{ deterministic} \end{cases}$$

Challenges

- Optimization is **NP-hard**
- Determining **feasibility** is NP-hard for > 1 constraint
- Standard approaches fail

Results

Assumption [TSR]: the cost of a policy is computable recursively over both **time** and state **space**

Theorem: Our algorithm is an **FPTAS** for any TSR criteria, which includes **expectation**, **almost-sure**, and **anytime** criteria

Remark: We only exclude chance constraints which are provably inapproximable

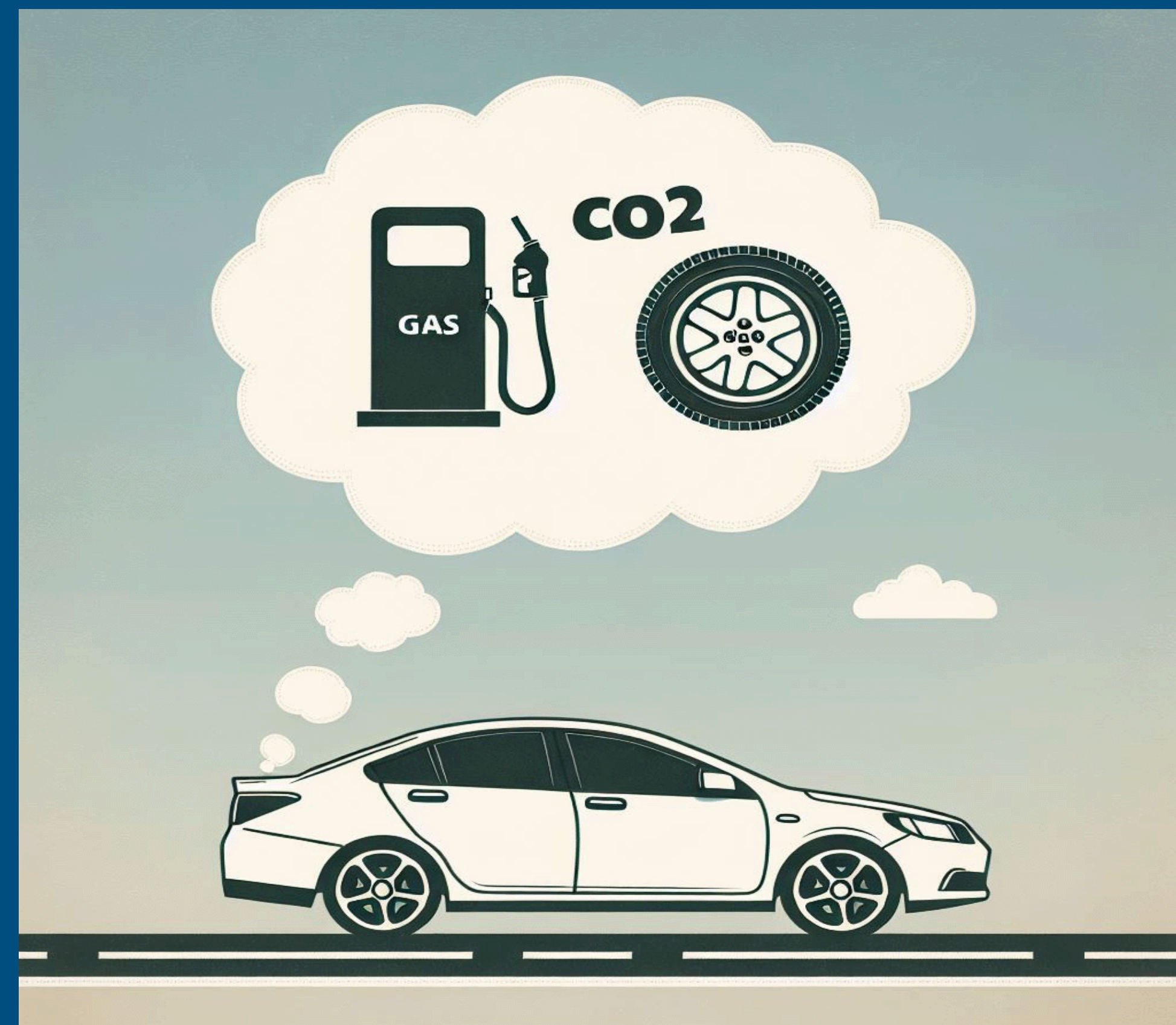
Conclusion

Polynomial-time **approximability is possible** for:

- Almost-sure constrained policies
- Anytime-constraint policies
- Expectation-constrained, deterministic policies

Was open for nearly 25 years!

Near-optimal **deterministic** policies for constrained RL can be computed **efficiently** for most constraint types.



Scan Me!

